CONFIGURATION OF NUT OPTIONAL WITHIN THE LIMITATIONS IMPOSED BY DIMENSIONS AND REQUIREMENTS SPECIFIED HEREIN

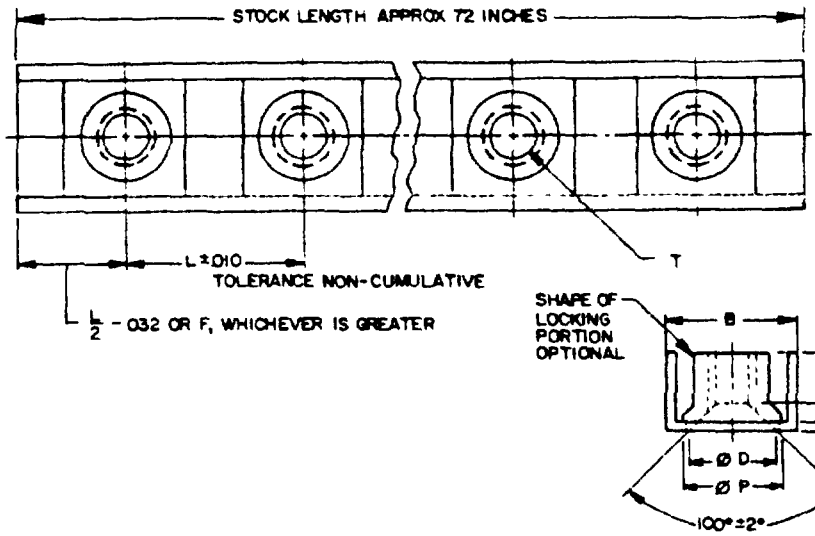


TABLE I DASH NUMBERS AND DIMENSIONS

FIRST DASH NUMBER 450° F	FIRST DASH NUMBER 800° F	THREAD SIZES	B MAX	Ø D MIN	F MIN	W MAX	L MIN SPACING AVAILABLE	H REF	Ø P² MAX	Ø P² MIN	X MAX	AXIAL STRENGTH LBS MIN	WEIGHT NUT ELEMENTS LBS .00	WEIGHT CHANNEL LBS .00
-05	-05	.64-32 UNJC-3B	.593	.360	.406	.27_	.750	.084	.41	.385	.031	.720	.24	.008
-	-	.90-24 UNJF-3B	.593	.406	.406	.28_	.750	.094	.45	.43	.0_	.466	.26	.008
-4	-4	.250-28 UNJF-3B	.874	.509	.56_	.14_	.000	.107	.62_	.560	.047	.4_6	.5_	.0_0

"H" MAX APPLIES TO NUT ELEMENT AND CHANNEL. MAX "H" NOT SPECIFIED IS LIMITED ONLY BY STRENGTH REQUIREMENTS OF SPECIFICATION.
HOLE IN CHANNEL MUST PROVIDE FOR FULL FLOAT OF NUT ELEMENT BUT NEED NOT BE CIRCULAR

TABLE II DASH NUMBERS AND DIMENSIONS

SECOND DASH NUMBER	L NUT ELEMENT SPACING	MAXIMUM NUMBER OF NUT ELEMENTS
-5	.625	115
-6	.750	96
-7	.875	82
-8	1.000	72
-9	1.125	64
-10	1.250	57
-11	1.375	52
-12	1.500	48
-13	1.625	44
-14	1.750	41
-15	1.875	38
-16	2.000	36
-18	2.250	32
-20	2.500	28
-24	3.000	24

Ⓕ ENTIRE STANDARD REVISED

PA NAVY-AS INTERNATIONAL INTEREST	TITLE NUT, SELF-LOCKING GANG CHANNEL, 100° CSK, FLOATING, LOW HEIGHT, CRES, 125KSI Ftu, 450°F & 800°F	MILITARY STANDARD	
Other Cust ARMY-AV AIR FORCE-11		MS21068	
PROCUREMENT SPECIFICATION M'L-N-25027	SUPERSEDES MS21089, NAS 1039 THRU 1041 IN PART	PAGE 1 OF 3	

DD 672-1 5310-1539

APPROVED 26 MAR 64 REVISED Ⓕ 30 SEP 87

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REQUIREMENTS

1 MATERIAL: NUT ELEMENT AND CHANNEL - CORROSION RESISTANT STEEL TYPE A286 (UNS S66286), IN ACCORDANCE WITH AMS5525, AMS5732 OR AMS5737 CHANNEL-TYPE 32, (UNS S32100), IN ACCORDANCE WITH AMS 5510

2 LENGTH SERVICES SHALL STOCK IN 72-INCH LENGTHS ONLY

3 FINISH ONLY 800° F NUTS SILVER PLATE ALL THREADED SURFACES OF NUT ELEMENT IN ACCORDANCE WITH AMS 2410 THREADS SHALL SHOW COMPLETE COVERAGE. SILVER PLATING ON ALL OTHER SURFACES OPTIONAL. TARNISHING OR DISCOLORATION OF SILVER NOT CAUSE FOR REJECTION CHANNEL - PLAIN (PASSIVATED)

4 LUBRICANT 450° F NUT ELEMENTS - DRY FILM LUBRICANT ON NUT ELEMENT ONLY, APPROVED IN ACCORDANCE WITH MIL-N-25027

5 DIMENSIONING AND TOLERANCING DIMENSIONING AND TOLERANCING SHALL BE IN ACCORDANCE WITH ANSI Y14 5M

6 HARDNESS NUT ELEMENTS SHALL HAVE A HARDNESS OF 49HRC, MAX

7 THREADS THREADS BEFORE LUBRICATION IN ACCORDANCE WITH MIL-S-8879

8 SURFACE TEXTURE SURFACE TEXTURE, UNLESS OTHERWISE SPECIFIED, SHALL NOT EXCEED 125 MICROINCHES, IN ACCORDANCE WITH ANSI B46.1.

9 PERFORMANCE SEE PROCUREMENT SPECIFICATION, EXCEPT 450°F DRY FILM LUBRICATED NUTS LIMITED TO FIVE REUSE CYCLES

10 FLOAT OF NUT ELEMENT FLOAT OF NUT ELEMENT PORTION OF ASSEMBLY SHALL NOT BE LESS THAN .030 NOR MORE THAN .040 LONGITUDINALLY AND NOT LESS THAN .010 NOR MORE THAN .030 LATERALLY FROM CENTERED POSITION NUT ELEMENT SHALL BE CAPABLE OF ENGAGEMENT WITH A BOLT IN THE MAXIMUM MISALIGNMENT POSITION MAXIMUM AXIAL FLOAT .020 INCH FOR -08 AND -3, .030 FOR -4

11 ASSEMBLY THE ASSEMBLY SHALL PROVIDE A BEARING SURFACE FOR THE NUT ELEMENT WITHIN THE HOUSING THE CENTERLINE OF THE CHANNEL SHALL NOT DEVIATE FROM A STRAIGHT LINE BY MORE THAN .015 IN ANY 12 INCHES THE NUT ELEMENT AND BASE PORTION OF THE ASSEMBLY SHALL FORM ONE INTEGRAL UNIT

12 EDGES BREAK SHARP CORNERS AND REMOVE ALL BURRS

13 CODE FIRST DASH NUMBER DESIGNATES THREAD SIZE.
SECOND DASH NUMBER DESIGNATES NUT ELEMENT SPACING IN EIGHTHS OF AN INCH
THIRD DASH NUMBER INDICATES NUMBER OF NUT ELEMENTS WHEN LESS THAN STOCK LENGTH (APPROX 72 INCHES) IS DESIRED
LETTER "L" BEFORE FIRST DASH NUMBER DESIGNATES DRY FILM LUBRICATED NUT ELEMENTS

14 PART NUMBER THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I AND TABLE II AND A THIRD DASH NUMBER, IF SPECIFIED (SEE REQUIREMENT 13)

EXAMPLE MS21068-3-6

- DASH NUMBER FROM TABLE II
- DASH NUMBER FROM TABLE I
- BASIC MS NUMBER

MS21068-3-6 INDICATES NUT, SELF-LOCKING GANG CHANNEL, 100° CSK, FLOATING, LOW HEIGHT, CRES 125 KSI Ftu, 800° F, GANG CHANNEL ASSEMBLY CONSISTING OF NINETY-SIX 190-32 UNJF-3B, SILVER PLATED NUT ELEMENTS SPACED AT .750 INCH, 800° F

MS21068L4-7-10 INDICATES NUT, SELF-LOCKING GANG CHANNEL, 100° CSK, FLOATING LOW HEIGHT CRES, 125 KSI Ftu, 450° F, GANG CHANNEL ASSEMBLY CONSISTING OF TEN .250-28 UNJF-3B, DRY FILM LUBRICATED NUT ELEMENTS, SPACED AT .875 INCH, 450° F

APPROVED 26 MAR 64 REVISED (F) FOR CHANGES SEE PAGES 1,2 AND 3

P A NAVY-AS Other Cust ARMY-AV AIR FORCE-11	INTERNATIONAL INTEREST	TITLE NUT, SELF-LOCKING GANG CHANNEL, 100° CSK, FLOATING, LOW HEIGHT, CRES, 125 KSI Ftu, 450°F & 800°F	MILITARY STANDARD MS21068
PROCUREMENT SPECIFICATION MIL-N-25027	SUPERSEDES MS21089, NAS1039 THRU 1041 IN PART		PAGE 2 OF 3

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES. DIMENSIONS TO BE MET PRIOR TO LUBRICATION.
2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.
4. DESIGN AND USAGE LIMITATIONS. THESE NUTS ARE DESIGNED TO DEVELOP THE TENSILE STRENGTH OF BOLTS AND SCREWS WITH AN ULTIMATE TENSILE STRENGTH OF 125 KSI BASED ON THE CROSS SECTION AREA AT THE BASIC ROOT DIAMETER OF THE THREADS. THESE NUTS ARE DESIGNED TO BE USED ON 3A EXTERNAL THREADS. THESE NUTS SHALL BE USED IN ACCORDANCE WITH THE LIMITATIONS OF MS33588. ONLY NUTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL 25027 SHALL BE USED.

INTERCHANGEABILITY TABLE

CANCELLED PART NUMBERS 1/	SUBSTITUTIVE PART NUMBERS 1/
NAS1039C	MS21068-08
NAS1040C	MS21068-3
NAS1041C	MS21068-4
MS21089L08	MS21068L08
MS21089L3	MS21068L3
MS21089L4	MS21068L4

1/ NUT ELEMENT SPACING DASH NUMBER IS IDENTICAL FOR BOTH CANCELLED AND CORRESPONDING SUBSTITUTIVE PART NUMBERS.

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